National Research University Higher School of Economics (HSE)

Curriculum
Field of study 01.04.02 Applied Mathematics and Informatics
Educational Programme "Math of Machine Learning"
Trajectories: "Math of Machine Learning"
Implementing unit: Faculty of Computer Science, HSE - Moscow
2 nd, 2026/2027 academic year

APPROVED
09.04.2026
Vice Rector
ROSHCHIN S.Y.
Signed with EDS

Length of Programme: 2 years
Years of Study: 2025/2026 - 2026/2027
Mode of Study: Full Time
Degree: Master's degree / MBA

Block Code	Course	Subject type	Department	Period of execution	Credits	Total Academic Hours	Contact Hours	Allocation of Contact Hours				Additional Information
								1	2	3	4	
	Degree Programme				**60,00**	**2 280**	**244**	**100**	**128**	**12**	**4**	
	Math of Machine Learning (General track)				**60,00**	**2 280**	**244**	**100**	**128**	**12**	**4**	
	Key Seminars				**9,00**	**342**	**44**	**16**	**16**	**12**		
1	Mentor's Seminar "Math of Machine Learning "	C	Department of Big Data and Information Retrieval		9,00	342	44	16	16	12A		Foreign language
	Final State Certification (FSC)				**6,00**	**228**	**2**				**2**	
1	Final Qualification Paper	C			6,00	228	2				2A	Foreign language
	Internship				**24,00**	**912**	**2**				**2**	
	Preparation of the Graduate Thesis				**24,00**	**912**	**2**				**2**	
1	Graduation Thesis	C		F	24,00	912	2				2A	Foreign language
	Major				**21,00**	**798**	**196**	**84**	**112**			
	Basic Components				**3,00**	**114**	**28**		**28**			
1	Modern Algorithmical Optimization	C	Department of Big Data and Information Retrieval		3,00	114	28		28A			Foreign language
	Elective Courses. 2 year				**18,00**	**684**	**168**	**84**	**84**			
1	Generative Models in Machine Learning	E	Skolkovo Institute of Science and Technology		3,00	114	28	28A				Foreign language
2	High Performance Python Lab	E	Skolkovo Institute of Science and Technology		3,00	114	28		28A			Foreign language
3	Leadership for Innovators	E	Skolkovo Institute of Science and Technology		3,00	114	28	28A				Foreign language
4	Mathematical Foundations of Reinforcement learning	E	Department of Big Data and Information Retrieval		3,00	114	28		28A			Foreign language
5	Reinforcement Learning	E	Skolkovo Institute of Science and Technology		6,00	228	56	28	28A			Foreign language
6	Optimal Transport and Its Applications	E	Department of Data Analysis and Artificial Intelligence		3,00	114	40			40A		Foreign language
7	Planning Algorithms in Artificial Intelligence	E	Skolkovo Institute of Science and Technology		3,00	114	28		28A			Foreign language
8	Theoretical Methods of Deep Learning	E	Skolkovo Institute of Science and Technology		3,00	114	28		28A			Foreign language

Curriculum agreed:

Academic Supervisor	Naumov A.A.	13.02.2026
Dean	ARZHANTSEV I.V.	13.02.2026
Head of Centre for Educational Model Design	LEPESHKIN I.A.	08.04.2026

* Subject type:

Compulsory course	C
Elective course	E

Period of execution:

Fixed	F